



ENGM 178 Technology Assessment – Final Presentation

Grid Scale Energy Technology Solutions for the State Of Connecticut

TEAM 13 Chad Klaas | Muskan Singh | Sia Puthusseri

Under the guidance of: Mr. Andrew Herchek & Prof. Eric Bish





Connecticut Targets
300 MW grid-scale
energy storage by
2024 & 1,000 MW by
2031 (Public Act 21-53)

SITUATION



Progress towards
target is uncertain. A
clear strategy to assess
and mitigate
challenges is required.

COMPLICATION

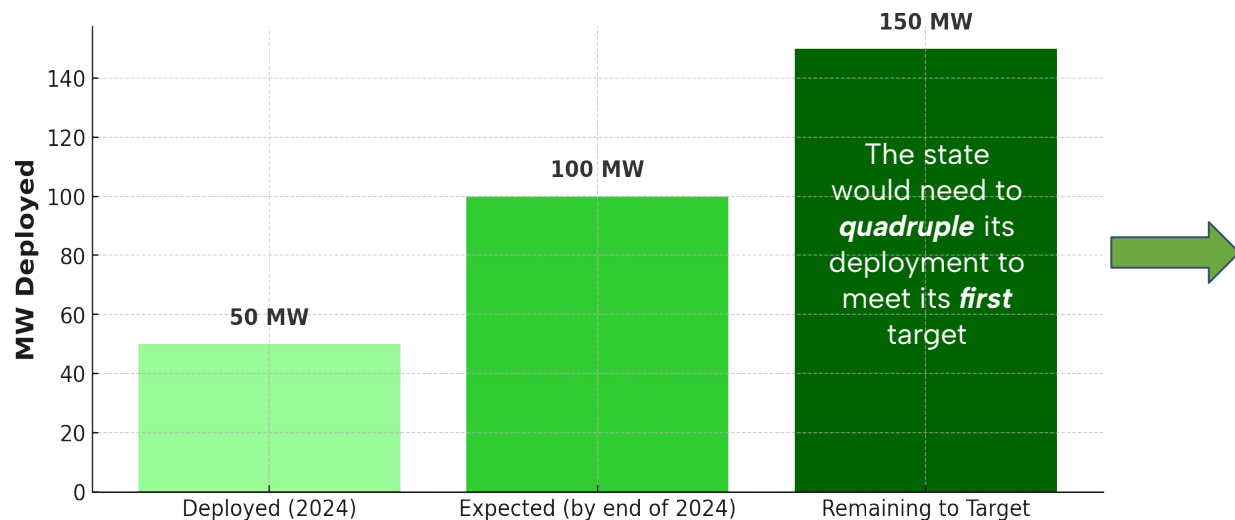


**Is Connecticut on
track for its 2024
storage target?
Roadmap for 1,000
MW by 2031**

KEY QUESTION

Connecticut is falling short by 150 MW To Meet Its Immediate Dec '24 Target Because Of Existing Infrastructure Challenges

Progress through Energy Storage Solutions (ESS) Program (MW Deployment)



Reasons they won't meet the target



LIMITED DEVELOPMENT



LACK OF SCALABILITY

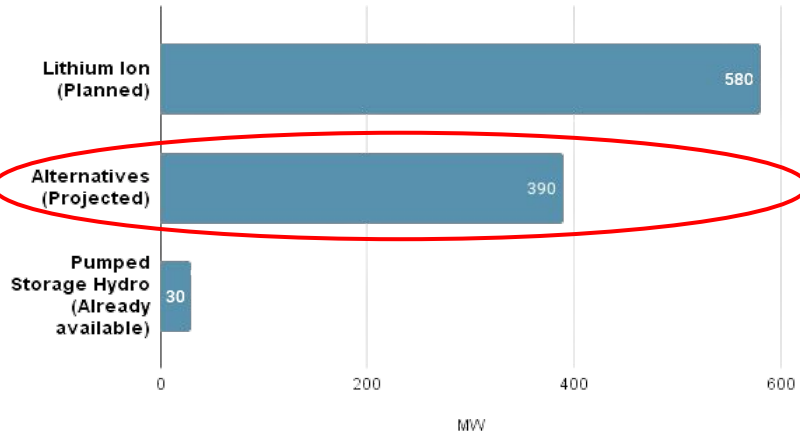


REGULATORY DELAYS

With their current plans, Connecticut Won't Meet Their 7-year Goal by a massive margin of 506 MW

390 MW

Failure to Match Alternatives' Target



116 MW

20% Reduction in Li-Ion



Environmental



Economic



Regulatory



Land Expansion



506 MW

Connecticut's Failure To Meet Targets Would Risk Economic, Environmental, and Grid Stability Impacts



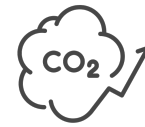
Grid Reliability & Security

Insufficient storage risks power shortages and outages impacting the population



Higher Energy Costs

Reliance on expensive alternatives will lead to budget wastage, causing economic capital wastage



Increased Carbon Emission

Reliance on fossil fuels will lead to severe environmental impacts

EXECUTIVE SUMMARY

Connecticut has not set the correct milestones for their 7-year energy roadmap



Li-Ion

150 MW

2024 - 2027
(Short Term)



Flow

700 MW

Dec 2031
(Long-Term)

Connecticut should continue their current use of **Li-ion batteries for 2024-27**, and perform a phase-shift to **Flow Batteries for 2031 and forward**

Connecticut Should Adjust Energy Storage Goal from 1000MW to 700MW by 2031 to Align with Available Resources

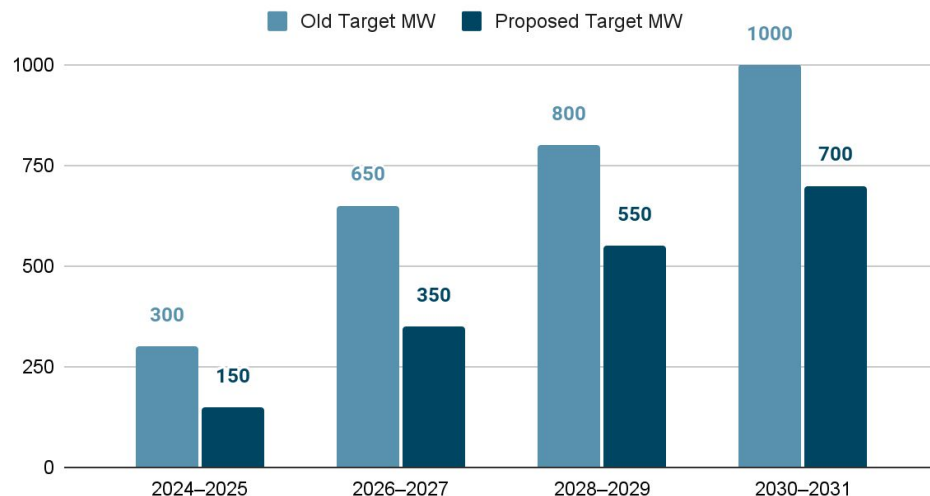
Current Rate:

- With the accounted for regulatory delays, the rate of energy deployment needed to meet 1000MW target is **150 MW/year** (unrealistic)

Proposed Milestones:

- Reducing the 2031 goal to 700MW and adjusting all milestones over 7 years would allow the adjusted required rate to be **75-100 MW/year (attainable)**

Old vs. New Proposed Storage Targets



Of All Storage Technologies Assessed (Used by Leading States), **Electrochemical** Works Best for Connecticut Given Its Topology, Policies, and Budget

Electrochemical Storage



Energy is stored in chemical reactions within batteries (e.g., lithium-ion, flow batteries)

Mechanical Storage



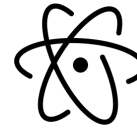
Stores energy through physical movement, such as pumped hydro or flywheels

Thermal Storage



Energy is stored as heat in materials (like molten salts) and released when needed*

Chemical Storage



Chemical storage stores energy through chemical reactions, like in batteries or hydrogen fuel cells*

Electromagnetic Storage



Energy is stored in magnetic fields, like in supercapacitors or superconducting magnetic energy storage (SMES)

* Technologies must prioritize safety and compliance with industry standards (Possibly explosive) or too R&D

Team Identified 3 Electrochemical Technologies Best Suited for Connecticut's Short and Long Term Energy Storage Goals

Li-Ion Batteries

Optimal for meeting immediate storage needs (up to 8 hours), fast deployment, and scalable

Flow Batteries

Best for 10+ hours of storage, scalable, and cost-effective and has potential for long-term solutions

Iron-Air Batteries

Promising for ultra-long-duration storage (100+ hours), still in RnD stage but could be ideal for long-term goals.

Primary and Secondary Research Led to Eight Criteria for Evaluating The 3 Prospective Technologies



Cycle Life

How long it will last i.e. number of charges



Round Trip Efficiency

Energy retained vs used (how effective)



Reliability

Degradation/Maintenance Needs Over Time



Cost Effective

Economical and budget friendly



Temperature Sensitivity

Performance variation related to temperature



Specific Energy

Energy stored per unit mass



Environmental

Minimal environmental impact



Safe

Minimal effects on health

Decision Matrix: Flow Batteries Outperform Li-Ion and Iron-Air in Key Metrics

Criteria	Weight	Lithium-Ion Batteries	Flow Batteries	Iron-Air Batteries
Cycle Life	14%	2	4	4
Temp Sensitivity	10%	2	4	3
Round-Trip Efficiency	13%	5	4	2
Specific Energy	8%	4	1	1
Safety	16%	3	4	5
Reliability	13%	3	4	4
Environmental	13%	2	4	5
Cost Effectiveness	13%	3	3	3
TOTAL	100%	24	28	27
WEIGHTED TOTAL		2.97	3.63	3.56

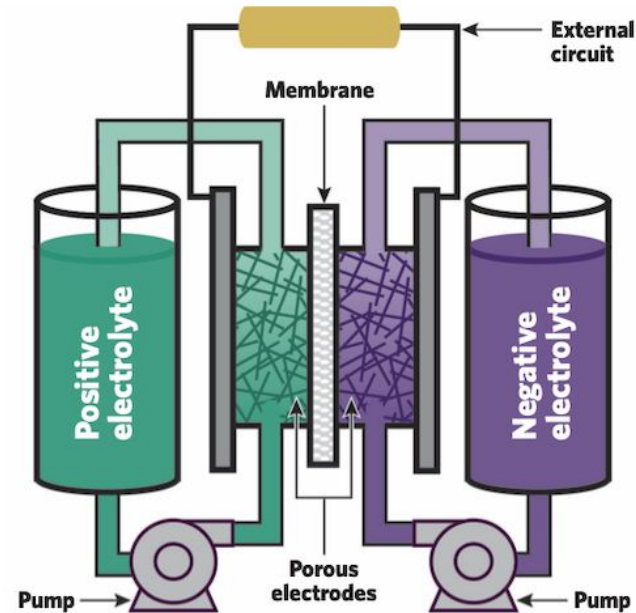
The score range is 1-5 where 5 represents the highest score

Weights for each metric computed using mean ratings by 2 subject matter experts

Flow Batteries Offer Longevity, Scalability, and Lower Lifetime Costs Despite Higher Initial Investment

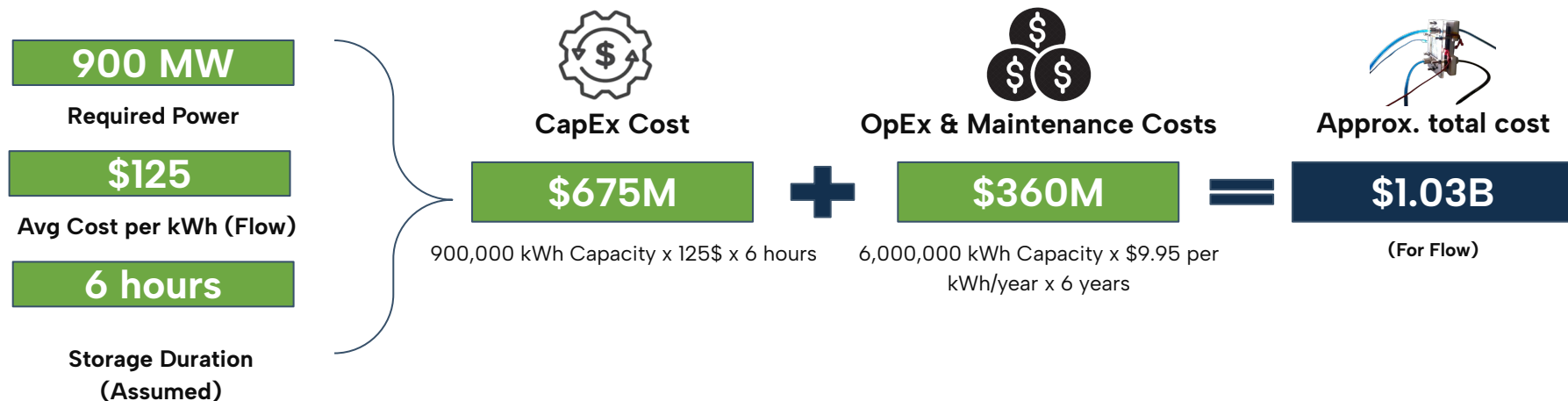
Flow batteries pump electrolytes through a reactor to convert chemical energy into electricity

- Long life cycle (25+ years)
- Low power density
- Easily scalable with tank size changes
- Higher upfront cost (\$750–\$4000) but lower lifetime cost
 - Tank costs vary
- Operates at lower temperatures



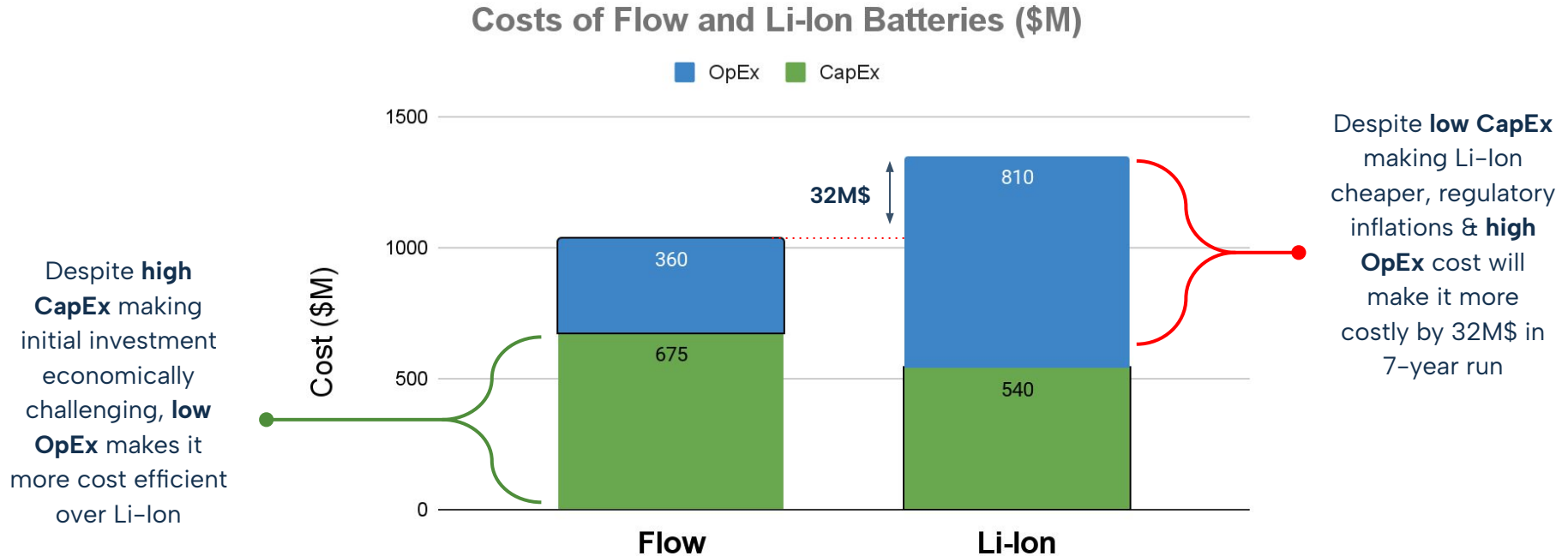


Connecticut Requires Roughly \$1B In Funding To Meet Its Energy Target Using Flow Batteries



But Is the transition from Li-Ion to flow batteries lucrative for Connecticut's 7-year plan?

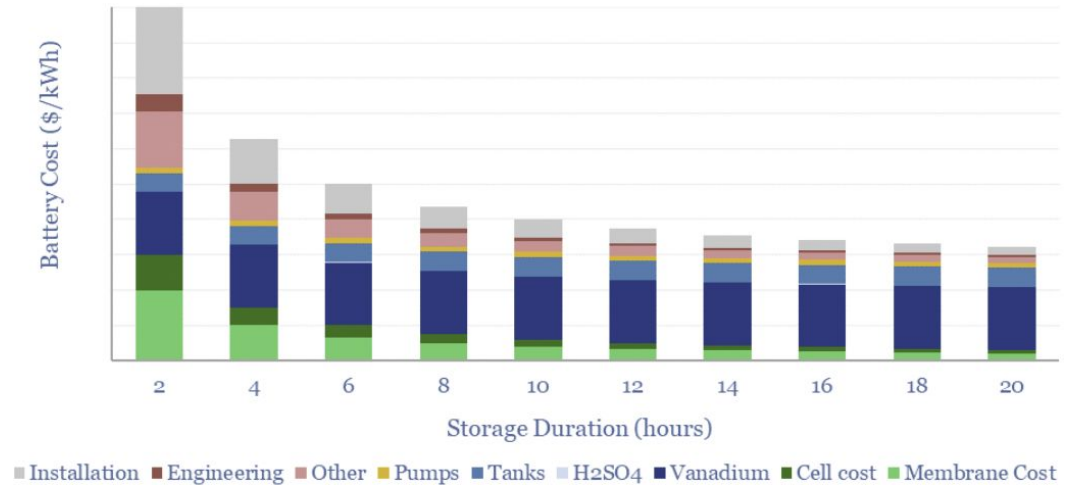
Flow Batteries Would Save Connecticut \$32M Compared to Li-Ion (Their Current Tech) by 2031 Despite Initially High Capex Cost



**Even Post 7 years, Flow Batteries Will Continue To Grow More Cost Efficient*

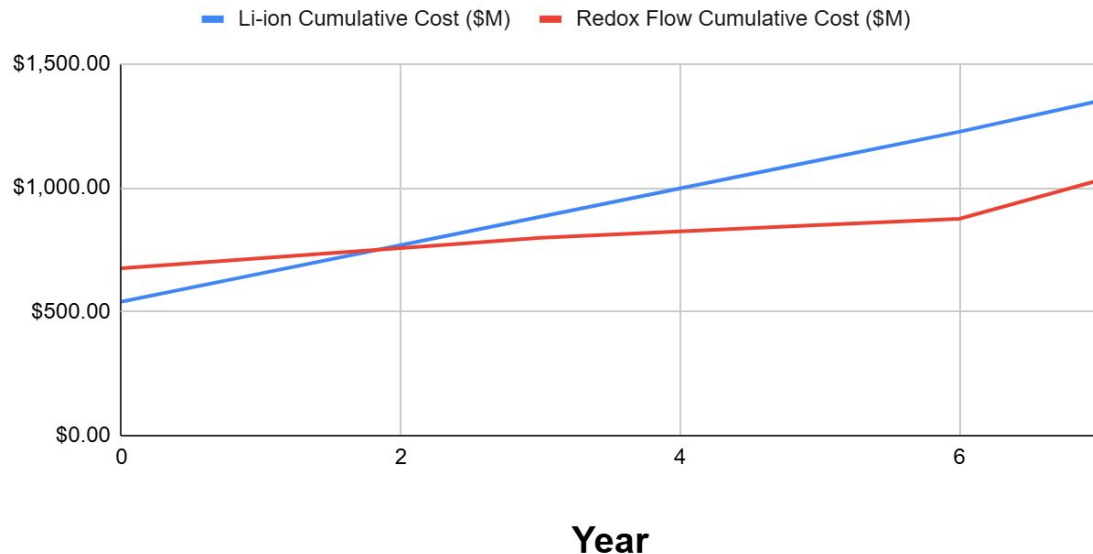
High CapEx Investment In Flow Will Eventually Lead To Massive Cost-Reductions In The Long-Run (20+ years)

- Flow Batteries have features that help in scaling up their capacity by enlarging the redox tanks
 - All other costs stay fixed
- A 12-hour flow battery charges and discharges 250 days per year will achieve same spread as 6-hour over 360 days
- Low cost to **continually increase** amount of energy stored and **reduce costs**



Year 3 Transition from Li-ion to Flow Batteries Is The Most Optimized Track To Achieve Adjusted 700MW Goal

When do Flow Batteries become more cost-efficient than Li-ion?



Annual Savings:

$$(\$1.35\text{B} - \$1.03\text{B}) / 7 = 45.71\text{M}$$

Payback Period:

$$= \text{CapEx Difference} / \text{Savings per year}$$

$$= (\$675\text{M} - \$540\text{M}) / \$45.71\text{M}$$

$$= 2 \text{ years (Starting around Year 3 Suggested)}$$

Until Year 3, Connecticut Should Focus On Establishing Flow Battery Distribution Channels (While Continuing Use Of Li-Ion)

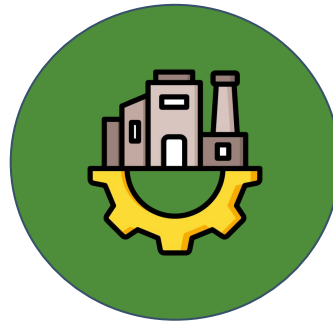
2025–2027: Continue Using Li-ion Batteries (Focus)

Goal: Build capacity and set the foundation for flow battery adoption.



Launch Residential & Commercial Programs

- **Target:** Homes & small businesses.
- **Action:** Incentive-driven installation with local contractors.



Expand to Industry & Municipalities

- **Target:** Factories & public buildings.
- **Action:** Showcase cost-saving potential & reliability for peak usage.



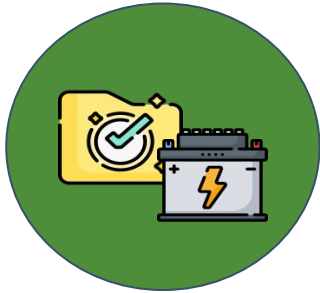
Integrate with Renewables

- **Target:** Solar/Wind Projects.
- **Action:** Position as a grid stability enhancer.

Post Year 3, Connecticut Should Entirely Transition to Flow Batteries For Achieving Remaining Target

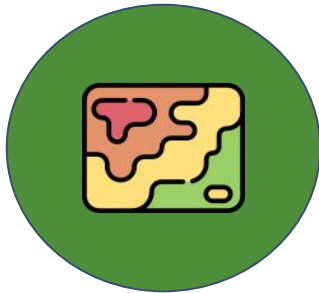
2028–2031: Transition to Flow Batteries

Goal: Introduce long-term, scalable energy storage.



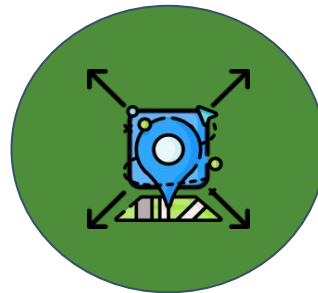
Pilot Flow Battery Projects

- **Target:** Factories & public buildings.
- **Action:** Showcase cost-saving potential & reliability for peak usage.



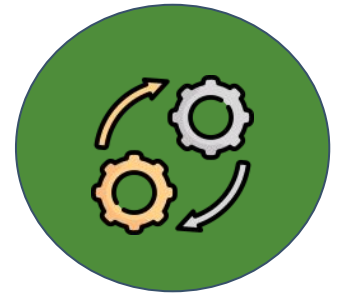
Overcome Topography Challenges

- **Target:** Identify suitable industrial/underutilized sites
- **Action:** Use suitable sites (e.g., decommissioned industrial zones).



Expand Statewide

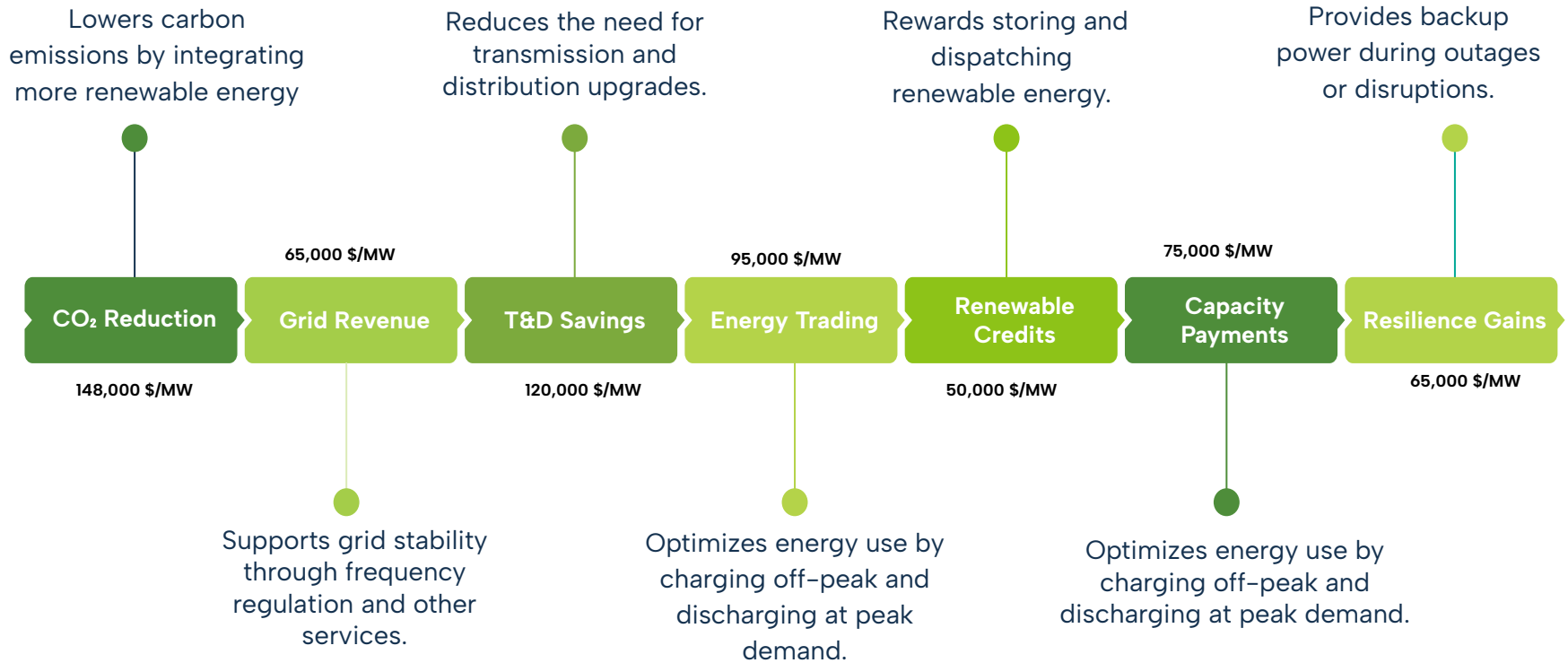
- **Target:** Commercial & grid operators.
- **Action:** Deploy at scale, leveraging state incentives.



Full Integration

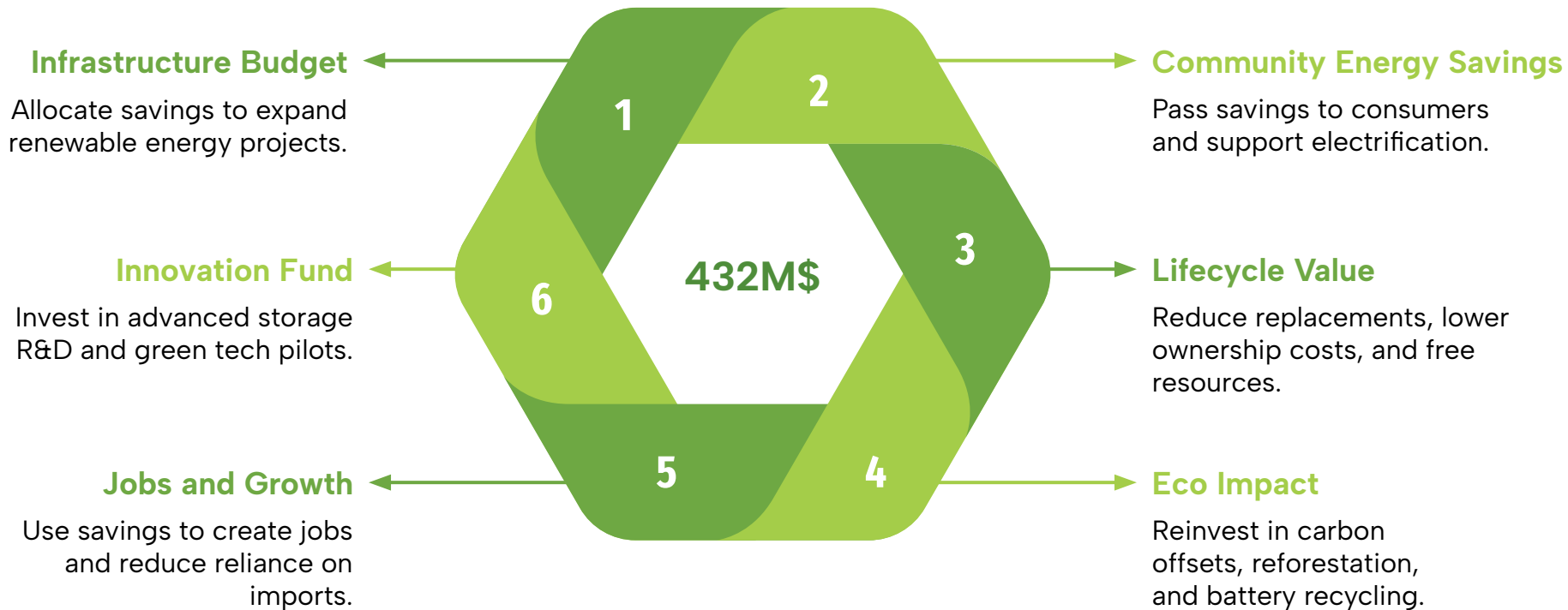
- **Target:** Collaborate with utilities and grid operators
- **Action:** Ensure smooth operation of flow batteries across Connecticut's grid.

Achieving 700 MW Target Through Flow Batteries Would Unlock a net Benefit of \$432M Across 7 Key Energy Revenue Streams



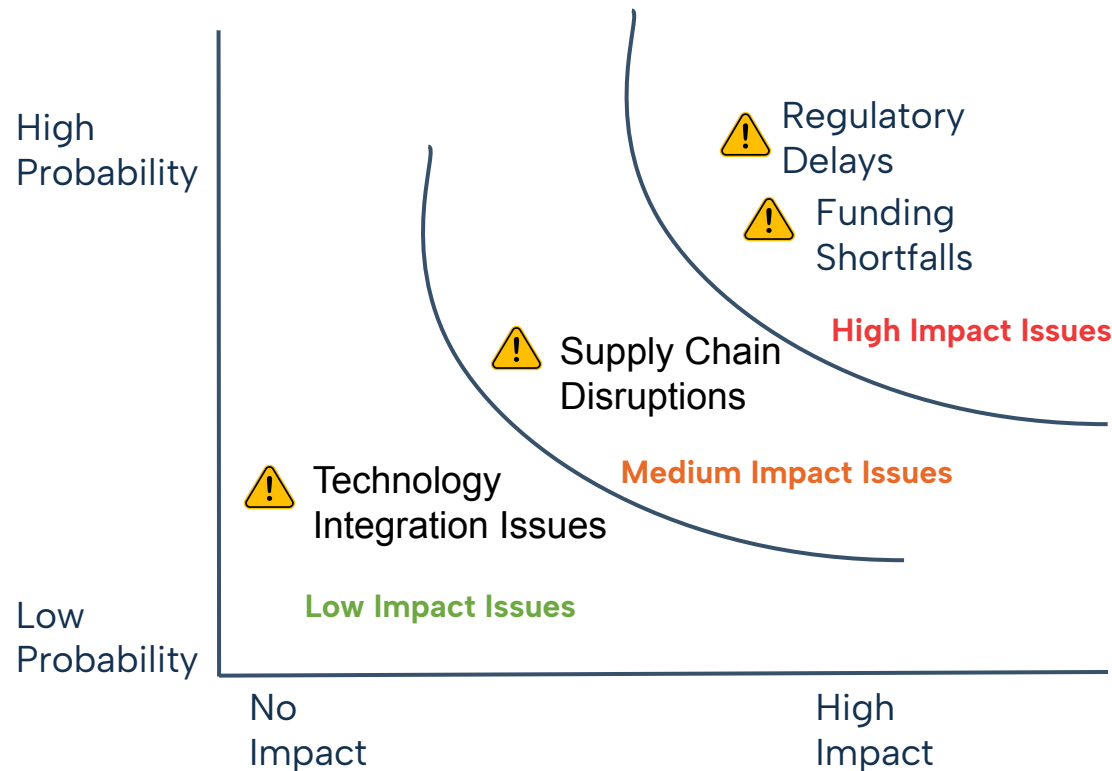
Benefit/MW = \$618,000/MW
Total Benefit = \$618,000 * 700 = \$432M

\$432M from Flow Batteries Would Fund More Scalable Energy Projects, Cost Savings, and Sustainability, Advancing Connecticut's Green Goals





Regulatory Delays & Funding Shortfalls are High Impact Risks In The Adoption Of Flow Batteries



RISK CATEGORIES

Low Impact Risks

Challenges in integrating flow batteries into Connecticut's existing grid infrastructure.

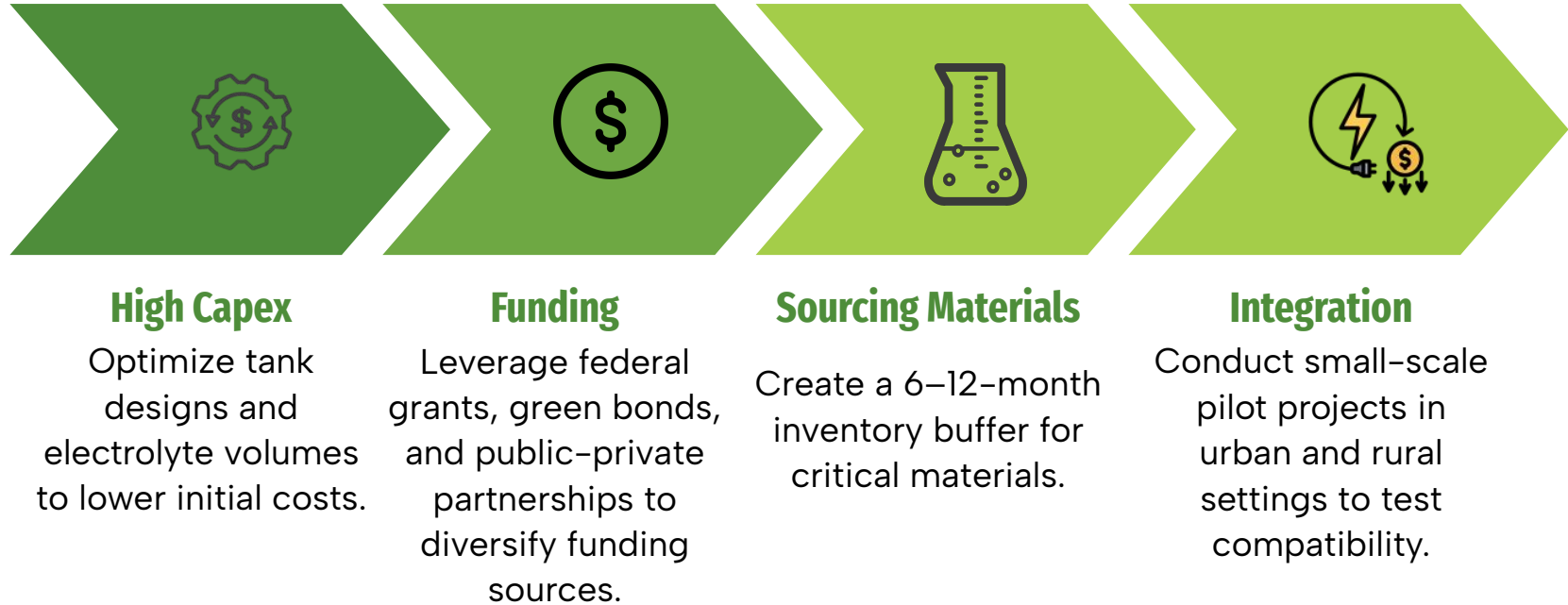
Medium Impact Risks

Disruptions in sourcing critical materials like vanadium or electrolyte components.

High Impact Risks

Connecticut's regulatory environment may slow approval for flow battery installations. Or Difficulty securing the \$1.03B required for flow battery adoption.

We evaluated mitigation strategies for the adoption of flow batteries in Connecticut through operational and economic precautions





Our Subject Matter Experts (SMEs) highlighted that adoption curve for new technologies like flow batteries should follow a phased trajectory



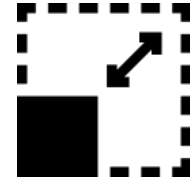
Early Adoption

Target pilot projects in specific regions, such as urban hubs like Hartford, or rural areas with renewable energy installations.



Infrastructure Upgrades

Gradual integration of flow battery systems into existing grid infrastructure, ensuring compatibility and resilience.



Scaling Deployment

Leveraging lessons from pilot projects to scale statewide, addressing both urban and rural needs.

**“It’s not just about how good the technology,
it’s also about whether it is the right time for the system to accept it.”**

~ Subject Matter Expert 1:
Dr. Azeez Mohammed
CEO of Covanta Energy

Recommended 7-Year Roadmap: Connecticut should use Li-Ion Batteries till 2027 and then shift to Flow Batteries after re-adjusting their goals down to 700 MW by 2031

Immediate Deployment (2024-2025)

Deploy **150 MW** using
Li-Ion batteries.



Optimization and Expansion (2028-2029)

Increase capacity to **550 MW**
with additional Flow Batteries.



Scaling Up (2026-2027)

Expand to **350 MW** with
Flow Battery integration.

Final Deployment and Evaluation (2030-2031)

Complete **700 MW** energy
storage capacity.



Connecticut's next steps should focus on revaluation of their existing goals, gathering necessary funds and launching pilot programs to diversify from Li-Ion

1. Reallocate Resources

As per suggested energy storage targets, refactor geographical, regulatory & other concerns



2. Diversifying Funding

Looking for funds outside main partner bank to expand energy programs of both Li-Ion & Flow wider



3. Map state areas

Assign Zones within the state to specific technologies on the basis of their individual needs



4. Pilot Projects

Locate necessary locations/facilities for testing alternative technologies



5. Training Skilled Labour

Retrain engineers and technicians to install and maintain upcoming technologies



Our Subject Matter Experts



Azeez Mohammed
CEO of Covanta Energy



Shailendra Waghulde
Products Specialist
(Siemens Limited.)

Thank You!

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Appendix

DETAILED EXECUTIVE SUMMARY

Is connecticut on track?

- **NO** Only **5.02 MW** deployed; requires **178 MW/year** to meet 2031 target because of high costs regulatory delays and scalability
- Key Challenges:
 - **High Costs:** \$2.35B for lithium-ion deployment is financially unfeasible.
 - **Regulatory Delays:** Approval processes and infrastructure readiness are major bottlenecks.
 - **Scalability Issues:** Limited projects and land-use constraints make rapid expansion unlikely.

Our Recommendation

- **Adjust Target:** Scale down the goal to a more feasible **700 MW by 2031**.
- **Adopt Flow Batteries:**
 - Long lifecycle (25+ years) with lower maintenance costs.
 - Scalable for Connecticut's needs.
 - Environmentally friendly, reducing rare material reliance.
 - Cost-effective over time despite upfront investment.



Flow

700 MW

Connecticut to Fall Further Behind Than Anticipated Due to Key Challenges of Its Star Technology – Li-Ion Batteries

Environmental Concerns



Water Usage
(Producing 1 ton of lithium consumes approx. 500,000 gallons of water) & risks associated to improper disposal and recycling

Economic Barriers



High Costs:
Advanced Li-Ion systems cost \$300–\$500 per kilowatt-hour. Very expensive to scale further as well.

Regulatory Challenges



Fast-evolving policies delay adoption of new technologies, increasing project costs by avg. 5–10%

Land Use & Expansion Limits

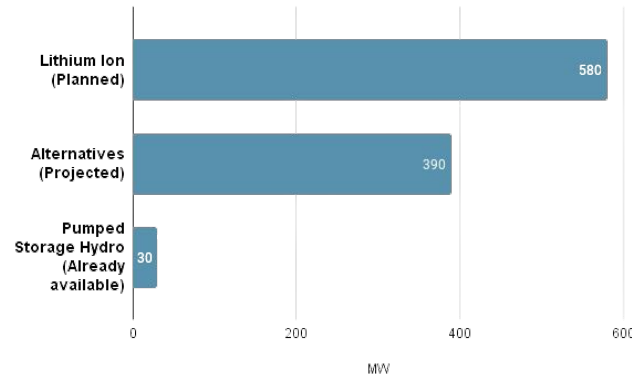


Connecticut lacks suitable topography for large-scale facilities and expansion of Li-Ion facilities requires large land areas

Factors cumulatively projected to cut Li-Ion batteries' viability by **15–20%**

6-year Energy Storage Plan Already Projecting a Minimum Shortfall of 390 MW

Current Plan intends to use mainly **Lithium-Ion batteries** to achieve 61% of 1000W target. Accounts for the remaining 39% as “alternative” (shortfall)



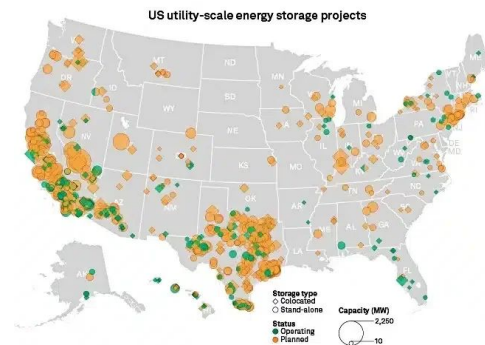
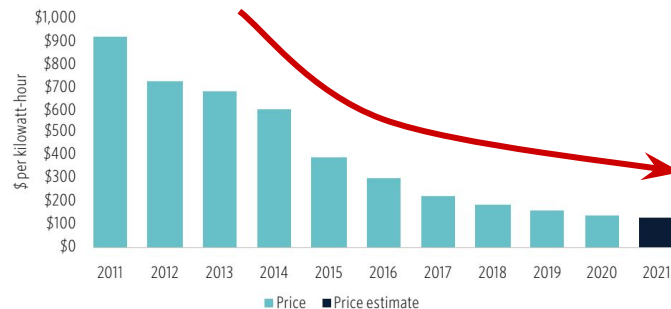
- 580 MW from planned battery systems (Lithium-Ion) through ESS program
- 30 MW already available from Pumped Storage Hydro facility at Rocky River
- 390 MW projected from additional battery systems or alternative technologies

Li-Ion Batteries Work Well For Current Short-Term Needs But Are Insufficient To Meet Long-Term Energy Storage Goals

Move lithium ions between an anode and a cathode creating electrical current

- 3-5 year life cycle, average of 3000 charging cycles
- Typically 85%–95% round trip efficiency, making it highly efficient for short-term use.
- Hold energy well for their mass and size (2-4 hour duration on a grid scale)
- Proven technology with decreasing costs due to mass production.
- Reliant on rare materials and negative environmental impact

Average pack price of Li-ion batteries*



Despite Projected Limitations of Li-Ion Battery Storage in Connecticut, Many U.S. States Using Similar Battery Technologies (Electrochemical) Are Leading the Way

1 California

8,179 MW of batteries

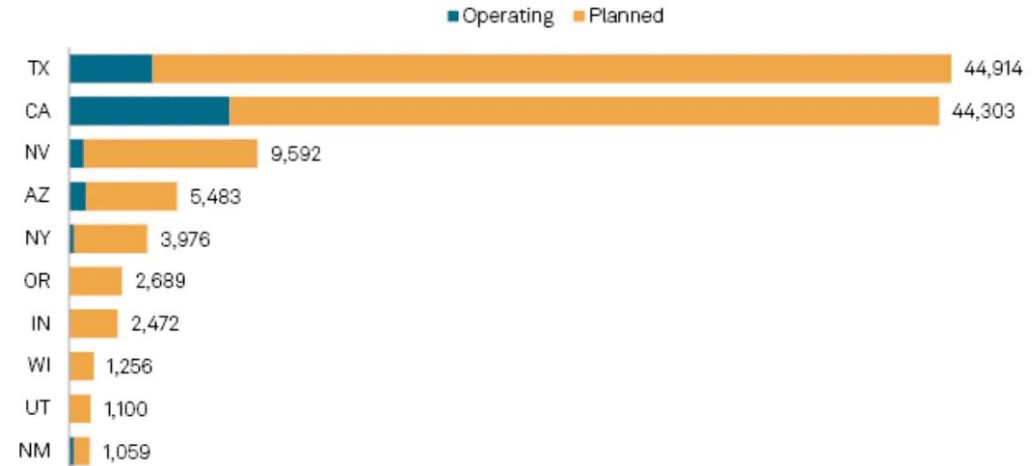
2 Texas

4,252 MW of batteries

3 Arizona

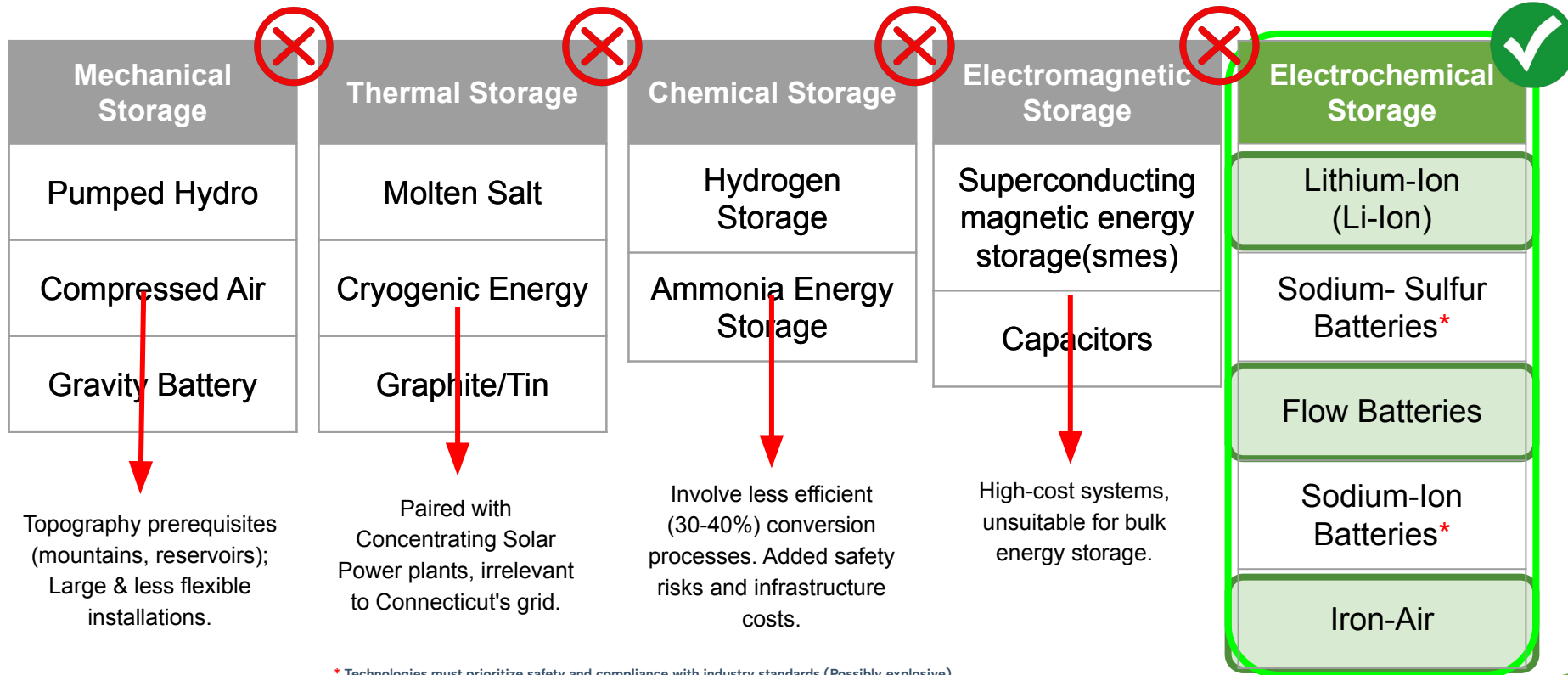
858 MW of batteries

States with largest utility-scale energy storage resources (MW)



OPPORTUNITY: *Connecticut still has ~100 MW of battery energy storage capacity untapped*

Of All Storage Technologies Used by Leading States, Electrochemical Works Best for Connecticut Given Its Topology, Policies, and Budget



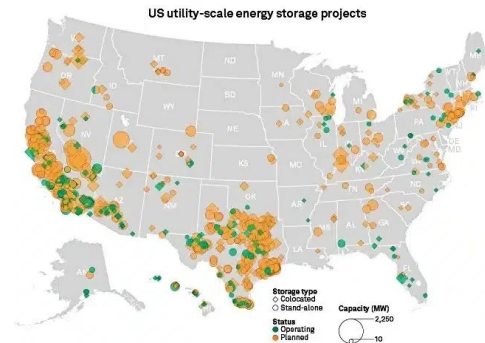
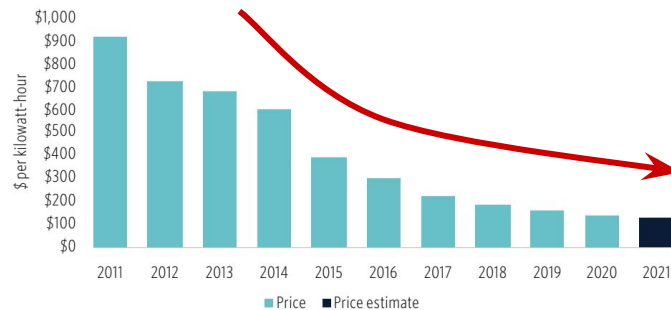
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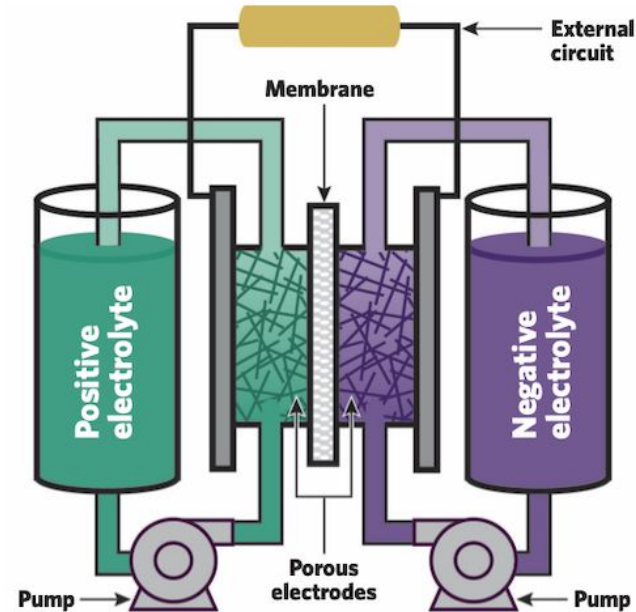
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While Flow Batteries Are Best for Long-Duration Storage

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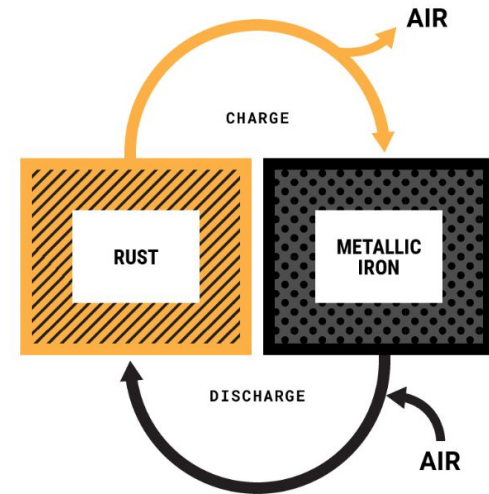
- Long life cycle (25+ years)
- Low power density
- Easily scalable with tank size changes
- Higher upfront cost (\$750-\$4000) but lower lifetime cost
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- Operates at lower temperatures



Iron-air Technology Could Be a Promising Long-Term Solution Following R&D

Offer 100+ hours of storage at lower costs than current technologies.

- Long life cycle (25+ years)
- Advantages:
 - Stores energy for 100+ hours.
 - Uses iron, making it cost-effective and sustainable.
- Challenges:
 - Still in development, needing more investment and testing for large-scale use.



The categorical benefits of Flow Batteries combined would net a benefit of \$493,000/MW per year

Category	Benefit (\$/MW)	Justification Notes
CO ₂ Reduction	148,000	Avoided 800 metric tons of CO ₂ emissions annually, valued at \$185/ton (Social Cost of Carbon).
Avoided T&D Costs	120,000	Reduced need for transmission and distribution upgrades.
Ancillary Services Revenue	65,000	Earnings from frequency regulation and other grid services.
Capacity Market Payments	75,000	Payments for ensuring grid reliability during peak demand.
Renewable Energy Credits	50,000	Credits for storing and dispatching renewable energy (valued at \$25/MWh for 2,000 MWh annually).
Resilience Benefits	65,000	Avoided costs of outages or power disruptions.
Market Arbitrage	95,000	Profits from buying off-peak energy and selling during peak demand.
TOTAL	\$618,000/MW	

The benefits of Flow Batteries Outweigh The Costs of transitioning from Li-Ion to Flow at 4-year benchmark in the roadmap

Total Costs: CapEx + OpEx = **\$1.03B**

Total Annual Benefits (7 Years) = *Annual Benefits (\$618M x 700)* = **\$432M**

$$\text{BCR} = 432,000,000 / 1,030,000,000 \approx \mathbf{0.42}$$

Low BCR in first 7 years: Project does not generate more monetary benefits than costs in the early years

Total Annual Benefits (17 Years) = *Annual Benefits (\$618M x 1700)* = **\$1.05B**

$$\text{BCR} = 1,050,000,000 / 1,030,000,000 \approx \mathbf{1.06}$$

Positive BCR after 17 years: Project will make a return on investment by year 17

The Payback Period of Flow Batteries is Within Connecticut's Timeline Despite The Apparent High Costs

Annual Revenue from Storage Spread:

- For **6-hour system**, using the **\$0.32/kWh** storage spread:
- **Total Annual Revenue for 6-hour system** = 2,160,000,000 kWh.

Total Annual Revenue (6-hour) = $\$0.32 \times 2,160,000,000 \text{ kWh} = \$691,200,000/\text{year}$.

Payback period = **CapEx / Annual Revenue** = $\$3,000,000,000 / \$691,200,000 = 4.34 \text{ years}$.

Qualitative Benefits of Flow Batteries

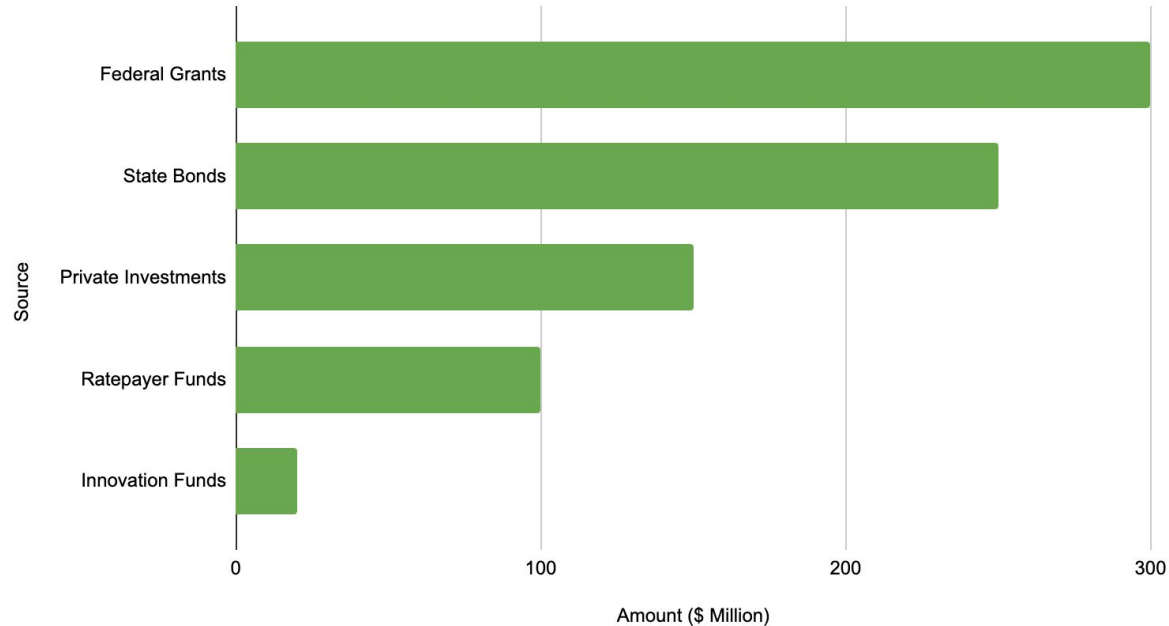
However, keeping in mind that these calculations do not factor in social benefits including...

- Higher GDP due to more consistent energy
- Healthier population
- Attracting more skilled labor

“Someone needs to be a leader for emerging technologies”

Market Analysis Of Funding Sources Backs The Viability Of Flow Batteries

Amount (\$ Million) vs. Source



\$840 Million

Total Estimated Funding across all sources

Various of these sources on top of the state's budget can help the state match the funding required for a new technology like Flow Batteries

(Estimated Cost: **\$1.03B**)

While Short Term Flow Batteries Have a Higher Costs, They Have a Significantly Lower Long Term Costs Compared to Li-Ion

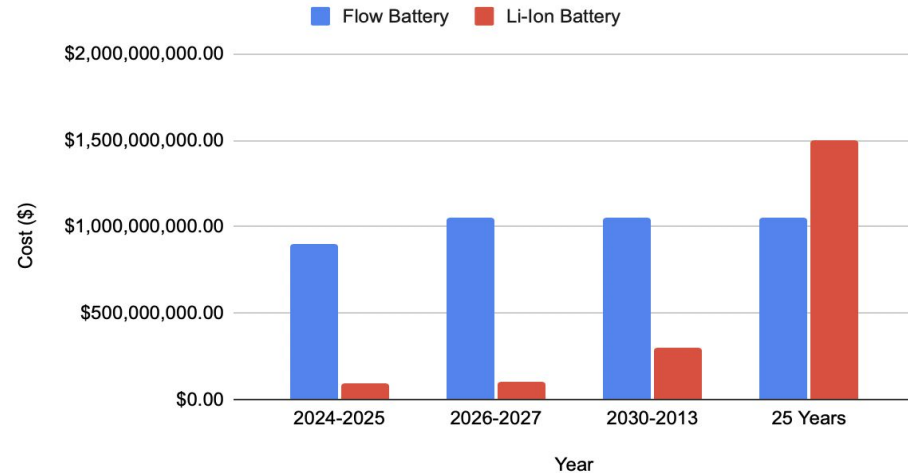
Total Costs: CapEx + OpEx = **\$1.03B**

Total Annual Benefits (7 Years) = \$432M

BCR ~ = 0.42

- High Upfront Costs
 - Infrastructure
 - Technology
- Gradual Benefit Realization:
 - Revenue from energy storage builds up incrementally

Cost Efficiency of Flow Batteries

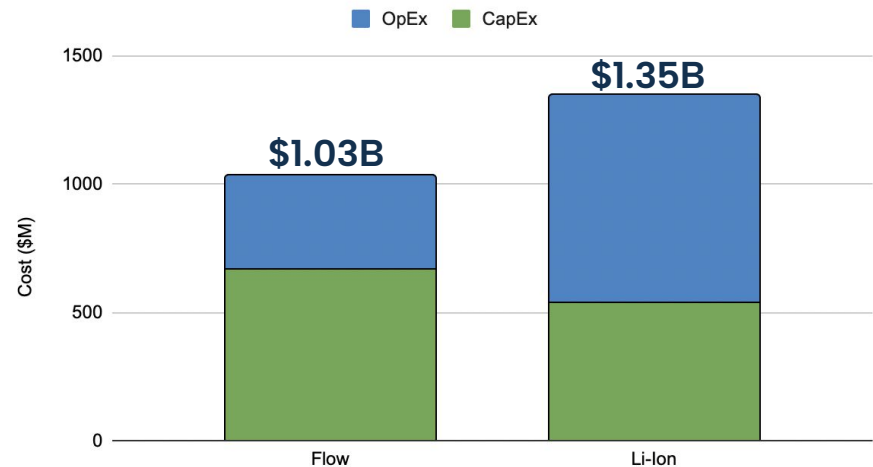


While **short-term monetary returns are limited**, the **long-term benefits** of consistent energy supply, grid reliability, and sustainability goals justify the investment

Flow Batteries Would Save Connecticut \$32M Compared to Li-Ion by 2031 Despite Initially High Capex Cost

- **Higher Initial CapEx for Flow Batteries:**
\$675M vs. \$540M
- **Total Savings:** Flow cost **\$1.03B** over vs. **\$1.35B** for 1000MW
 - Due to efficiency and durability of flow batteries...5x fewer replacements
- **Flow Battery Operation Costs:** \$9.95/kW-yr
- **Scalability:** Flow Batteries offer flexible capacity expansion with lower incremental costs over time

Costs of Flow and Li-Ion Batteries (\$M)



Overtime, Flow Batteries Will Become Immensely Cost Efficient

Decision Matrix – What 1–5 means?

Metric	1	2	3	4	5
Cycle Life (Number of Charge/Discharge Cycles)	Less than 500 cycles	500 to 2000 cycles	2000 to 5000 cycles	5000 to 10,000 cycles	More than 10,000 cycles
Temperature Sensitivity (Efficiency Drop in Extreme Temperatures)	Efficiency drop greater than 40%	Efficiency drop between 30-40%	Efficiency drop between 20-30%	Efficiency drop between 10-20%	Efficiency drop less than 10%
Round-Trip Efficiency (Percentage of Energy Retained During Charge/Discharge)	Less than 60% efficiency	60-70% efficiency	70-80% efficiency	80-90% efficiency	More than 90% efficiency
Specific Energy (Wh/kg - Energy Density by Weight)	Less than 50 Wh/kg	50-100 Wh/kg	100-150 Wh/kg	150-200 Wh/kg	More than 200 Wh/kg
Safety (Risk of Fire/Explosion and Toxicity)	High risk of fire, explosion, or environmental hazards under normal operating conditions	Moderate risk, requiring strict monitoring	Some risk but manageable with regular safety protocols	Minimal risk, no significant incidents in controlled conditions	No risk of fire, explosion, or environmental hazards under typical conditions
Reliability (Degradation, Maintenance Needs Over Time)	Rapid degradation or frequent breakdowns (e.g., after 1-2 years of use)	Degradation becomes noticeable within 2-4 years of use	Moderate degradation over 5-7 years, regular maintenance required	Minimal degradation over 7-10 years, occasional maintenance required	Very reliable for over 10 years, negligible degradation or need for maintenance
Environmental Impact (Material Sourcing, Toxicity, Recyclability)	Significant environmental and social costs (e.g., rare earth mining, hazardous waste, low recyclability)	High impact, with limited recycling infrastructure	Moderate impact, some issues with material sourcing or disposal	Low impact, recyclable materials are used but infrastructure is still developing	Very low impact, materials are abundant, non-toxic, and easily recyclable
Cost Effectiveness (Initial, Operational, and Maintenance Costs Over Time)	High upfront costs and high operational/maintenance costs, short lifespan	High upfront costs but moderate operational costs, short lifespan	Moderate upfront and operational costs, mid-level lifespan	Lower upfront costs and minimal operational/maintenance costs, long lifespan	Low upfront costs, minimal ongoing costs, long lifespan

Weights – Individual (How they were computed?)

Metric	Expert 1	Expert 2	Average	Avg/Summation(Avg)	Final Weights
Cycle Life (Number of Charge/Discharge Cycles)	4	5	$(4 + 5) / 2 = 4.5$	$4.5 / 31.5 \approx 0.143$	0.143
Temperature Sensitivity (Efficiency Drop in Extreme Temperatures)	3	4	$(3 + 4) / 2 = 3.5$	$3.5 / 31.5 \approx 0.111$	0.111
Round-Trip Efficiency (Percentage of Energy Retained During Charge/Discharge)	3	5	$(3 + 5) / 2 = 4.0$	$4.0 / 31.5 \approx 0.127$	0.127
Specific Energy (Wh/kg - Energy Density by Weight)	2	3	$(2 + 3) / 2 = 2.5$	$2.5 / 31.5 \approx 0.079$	0.079
Safety (Risk of Fire/Explosion and Toxicity)	5	5	$(5 + 5) / 2 = 5.0$	$5.0 / 31.5 \approx 0.159$	0.159
Reliability (Degradation, Maintenance Needs Over Time)	5	4	$(4 + 4) / 2 = 4.0$	$4.0 / 31.5 \approx 0.127$	0.127
Environmental Impact (Material Sourcing, Toxicity, Recyclability)	4	4	$(4 + 4) / 2 = 4.0$	$4.0 / 31.5 \approx 0.127$	0.127
Cost Effectiveness (Initial, Operational, and Maintenance Costs Over Time)	3	5	$(3 + 5) / 2 = 4.0$	$4.0 / 31.5 \approx 0.127$	0.127
		Sum	31.5		

Cycle Life:

- **Lithium-Ion (4)**: Solid cycle life but not the longest.
- **Flow Batteries (5)**: Superior cycle life, excelling in longevity.
- **Iron-Air (3)**: Shorter cycle life compared to others.

Temperature Sensitivity:

- **Lithium-Ion (3)**: Sensitive to temperature changes.
- **Flow Batteries (4)**: Moderate tolerance to temperature.
- **Iron-Air (5)**: Highly resistant to temperature fluctuations.

Round-Trip Efficiency:

- **Lithium-Ion (5)**: Excellent energy efficiency.
- **Flow Batteries (4)**: Good efficiency, but slightly lower than lithium-ion.
- **Iron-Air (3)**: Lower round-trip efficiency.

Specific Energy:

- **Lithium-Ion (4)**: High energy density.
- **Flow Batteries (3)**: Moderate energy density.
- **Iron-Air (2)**: Lower energy density.

Safety:

- **Lithium-Ion (3)**: Some safety concerns with overheating and flammability.
- **Flow Batteries (5)**: Very safe due to non-flammable materials.
- **Iron-Air (5)**: High safety profile with no major risks.

Reliability:

- **Lithium-Ion (4)**: Reliable but can degrade over time.
- **Flow Batteries (5)**: Very reliable with fewer performance issues.
- **Iron-Air (3)**: Reliability still improving but less proven.

Environmental Impact:

- **Lithium-Ion (3)**: Environmental impact from mining and disposal.
- **Flow Batteries (4)**: Less harmful materials, better recyclability.
- **Iron-Air (5)**: Uses abundant and environmentally friendly materials.

Cost Effectiveness:

- **Lithium-Ion (5)**: Economically viable with declining costs.
- **Flow Batteries (4)**: Slightly higher costs, but improving.
- **Iron-Air (4)**: Still emerging, but potential for cost reductions.

California and New York Succeeded by Leveraging Diverse Storage Solutions

California:

- Leveraged large-scale Li-Ion deployments for immediate storage needs.
- Shifting towards flow batteries and green hydrogen for long-duration storage.
- Helped manage peak demand and mitigate rolling blackouts

New York

- Focused on front-of-the-meter (FTM) storage and community battery projects.
- Initially relied on Li-Ion, now investing in flow batteries for longer-term storage.
- Aiming to install 6,000 MW by 2030, emphasizing diverse technologies

Emerging Technologies are Critical to Closing Connecticut's Storage Gap

Battery type	Advantages	Disadvantages
Li-ion	• High maturity of technology, manufacturing, and supply chains. • High energy density. • Low overall cost. • Cells from high-performance applications, such as EVs, can be repurposed as BESSs.	• High cost to scale storage duration. • Dependent on rare natural resources for electrolyte and cathodes. • Flammable materials.
Redox flow	• Simple, low-cost scaling through increasing electrolyte storage tanks. • Long lifespan for installations, performance does not degrade for 25 to 30 years.¹¹ • Inherent fire safety—non-flammable energy storage medium. • Not reliant on rare resources or those with problematic supply chains • Energy storage capacity (electrolyte tank size) uncoupled from power output (electrochemical cell size), simplifying tailored installations.	• Low energy density, not well suited to mobile applications, thus limiting shared research and development. • Low technological maturity relative to Li-ion, hindering effective evaluation of performance.¹² • High costs for short duration installations. • Relatively low numbers of installations, complicating assessments of safety and maintenance.
Metal-air	• High energy density—such as for zinc-air batteries. • Uses low-cost, abundant raw materials. • Inherent fire safety—non-flammable storage medium. • Cost to scale duration lower than Li-ion.	• Low energy density (for iron-air batteries). • Some challenges to be mitigated in charging/discharging. • Relatively low numbers of installations, complicating assessments of safety and maintenance.

Mechanical Storage

- **Geographical Constraints:** PSH needs specific topography (mountains, reservoirs), unavailable in Connecticut .
- **Infrastructure Needs:** CAES and gravity systems require large installations, less flexible than modular battery systems .
- **Energy Release:** Primarily suited for short-term, rapid energy release .

Thermal Storage

- **Use Case Mismatch:** Technologies like molten salt storage are paired with CSP plants, irrelevant to Connecticut's grid .
- **Efficiency:** Lower round-trip efficiency (~50%) compared to electrochemical systems .

Chemical Storage

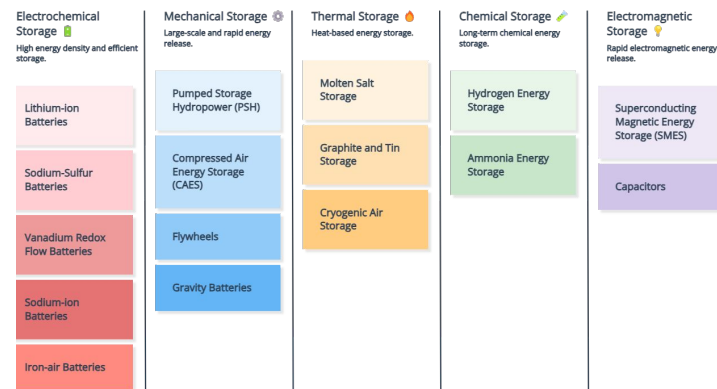
- **Conversion Efficiency:** Hydrogen and ammonia storage involve complex, less efficient (30-40%) conversion processes .
- **Infrastructure:** Requires new pipelines and tanks, posing safety risks and infrastructure costs .

Electromagnetic Storage

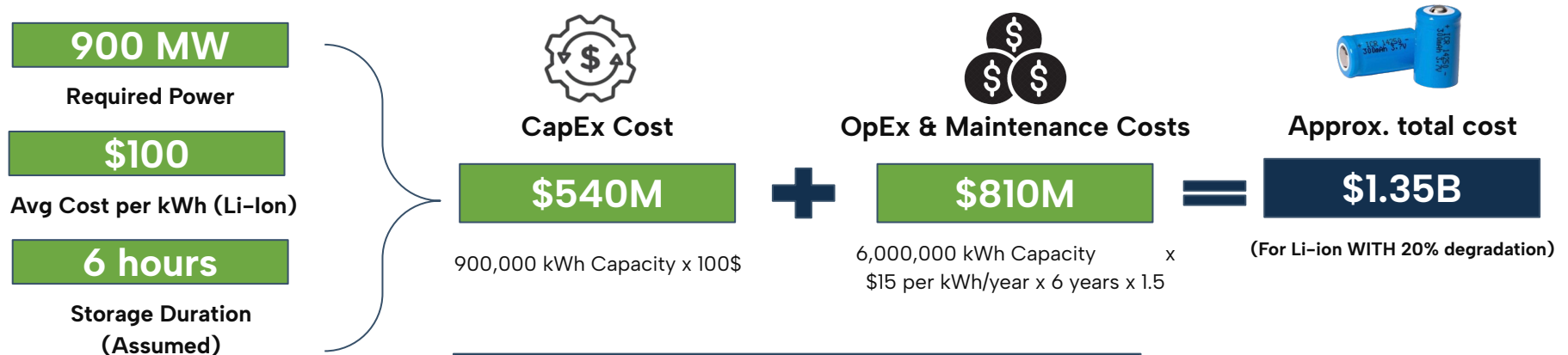
- **Energy Capacity:** Capacitors and SMES are great for seconds-long energy bursts, not grid-scale storage .
- **Cost:** High-cost systems unsuitable for bulk energy storage .

Focusing on Electrochemical Storage

- **Scalability:** Lithium-ion batteries are scalable, modular, and cost-effective .
- **Efficiency:** Electrochemical systems provide 80-95% efficiency .
- **Cost:** Battery costs have dropped by 85% in the last decade .



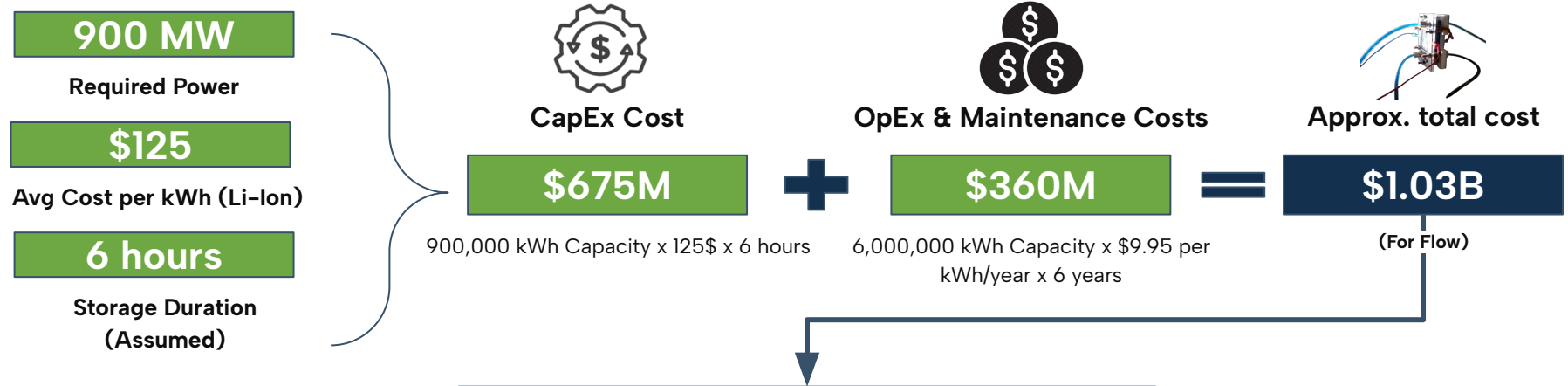
When Cost Of Li-Ion \$1.03B shows its financially feasible to use them for Connecticut's 1000MW goal based on other state benchmarks



Li-Ion Batteries: Cost Evaluation

Flow Batteries will be cheaper as compared to Li-Ion by 32M in the 7-year run

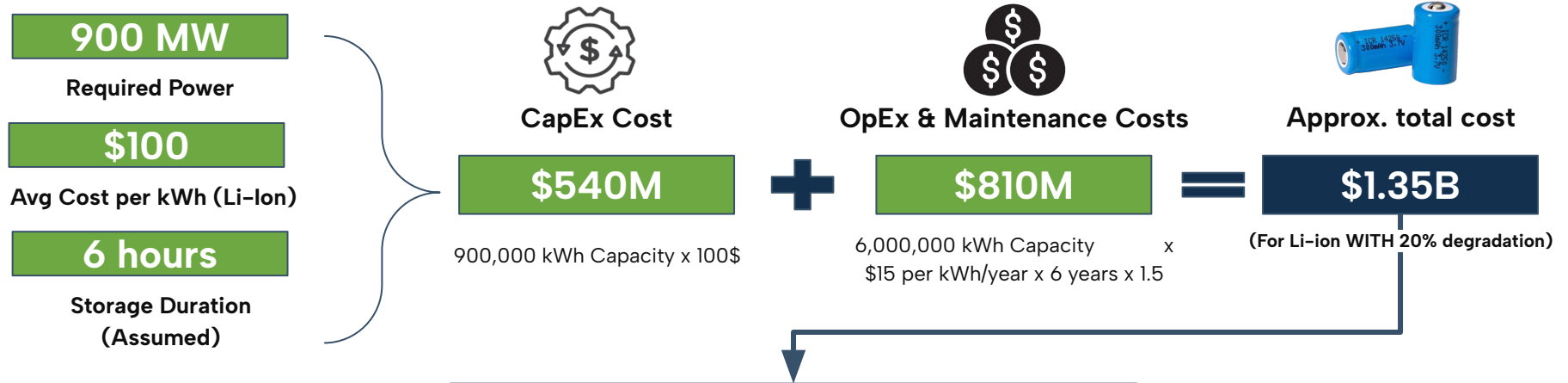
Calculating The Cost Of Flow Batteries \$1.03B shows its financially feasible to use them for Connecticut's 1000MW goal



Flow Batteries: Cost Evaluation

The cost of 7-year Flow Batteries Plan for Connecticut is in-budget compared to successful states like California (\$1.2B) & Texas (\$7B)

When Cost Of Li-Ion \$1.03B shows its financially feasible to use them for Connecticut's 1000MW goal based on other state benchmarks



Li-Ion Batteries: Cost Evaluation

Flow Batteries will be cheaper as compared to Li-Ion by 32M in the 7-year run

Li-Ion is cheaper to set up but has a higher OpEx cost + recharges 1.5 times more frequently than Flow batteries, making them costlier than Flow by 32M in the 7-year run

Left Energy Storage Goal: $1000 - 100 = 900$ MW

- 6-hour redox flow battery: \$100/kWh per hour
- Capital Expenditure (CapEx) = $900,000 \text{ kWh} \times 6 \times 100\$ = \mathbf{\$540M}$
- Operational Expenditure (OpEx) = $6,000,000 \text{ kWh} \times (15\$/\text{kWh}/\text{year}) \times 6 \times 1.5 = \mathbf{\$810M}$
- **Total = \$1.35B**

CALCULATION:

OPEX-

Capacity (kWh)= >

Power (MW) × Storage Duration (hours)

$1000\text{MW} \times 6\text{hours} = 6000 \text{ MWh} = 6,000,000 \text{ kWh}$

CAPEX- Left over goal - 900 MW => Capacity = Power (MW) x Storage Duration = $900,000 \times 6 \times 125$

The benefits of Flow Batteries Outweigh The Costs of transitioning from Li-Ion to Flow at 4-year benchmark in the roadmap

Total Annual Benefits:

Total Annual Benefits for 7 Years = Annual Benefits (618x700) = \$432 Million

Total Costs: CapEx Cost + OpEx Cost = \$1.03 Billion

Benefit-Cost Ratio (BCR):

BCR = Total Benefits / Total Costs

BCR = 432,000,000 / 1,030,000,000 $\approx$ **0.42**

Quantitatively – show BCR as low for 7 years

After that, show BCR > 1 calc after 17 years = 1.02

This shows that purely on monetary benefits, project will make a return after 16 years.

However, keeping in mind that these calculations do not factor in social benefits. Such as higher gdp due to more consistent energy, a more healthy & content population, attracting higher skilled labor

Additional Info to add to Slide 3 – Flow batteries have high CapEx but low OpEx, making them a cost-effective for long-duration (6+ hours) energy solution over-time despite high CapEx

Capital Expenditures (CapEx)

- Electrolyte Solution: \$200/kWh
- Cell Stacks: \$250/kWh
- Storage Tanks: \$150/kWh
- Power Conversion Systems: \$100/kWh
- Balance of Plant: \$50/kWh
- **Total: \$750/kWh**
(750/6= 125\$/kWh per hour)

Operational Expenditures (OpEx)

- Routine Maintenance
- System Monitoring
- Electrolyte Replenishment
- Component Replacements
- **Total: \$9.95/kWh/year**

Economic Capital Costs for Flow

Target Energy Storage Goal: 1000MW

- 6-hour redox flow battery: \$3000/kWh
- Energy Usage Per Year:

$$1,000,000 \text{ kW} \times 360 \text{ days/year} \times 6 \text{ hours/day} = \mathbf{2,160,000,000 \text{ kW/yr}}$$

- Capital Expenditure (CapEx)

$$1,000,000 \text{ kW} \times \$3000/\text{kW} = \mathbf{\$3,000,000,000}$$

Appendix Slide

Left Energy Storage Goal: $1000 - 100 = 900$ MW

- 6-hour redox flow battery: \$125/kWh per hour
- Capital Expenditure (CapEx) = $900,000 \text{ kWh} \times 6 \times 125\$ = \mathbf{\$675M}$
- Operational Expenditure (OpEx) = $6,000,000 \text{ kWh} \times 9.95\$/\text{kWh}/\text{year} \times 6 \text{ years} = \mathbf{\$358.2M}$
- **Total = \$1.03B**

CALCULATION

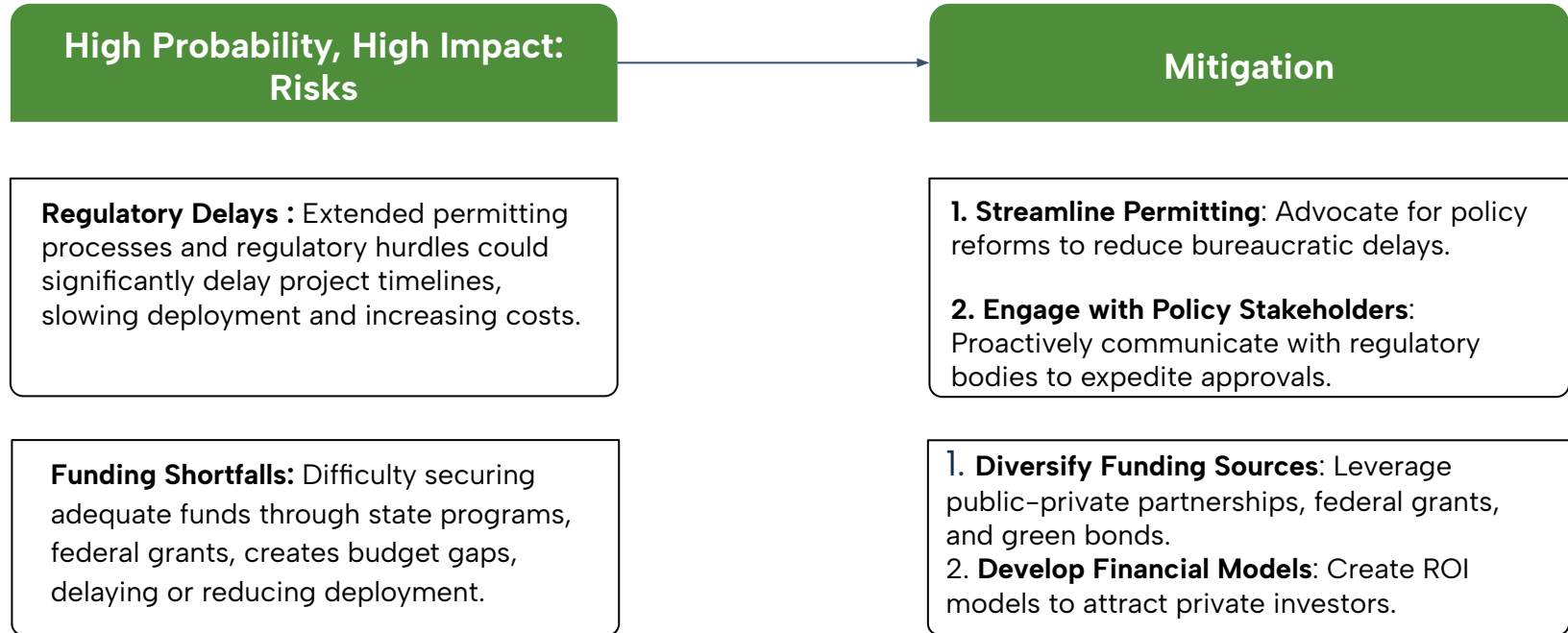
OPEX

=> Capacity (kWh) = Power (MW) $\times$ Storage Duration (hours) = $1000\text{MW} \times 6 \text{ hrs}$
= $6000 \text{ MWh} = 6,000,000 \text{ kWh}$

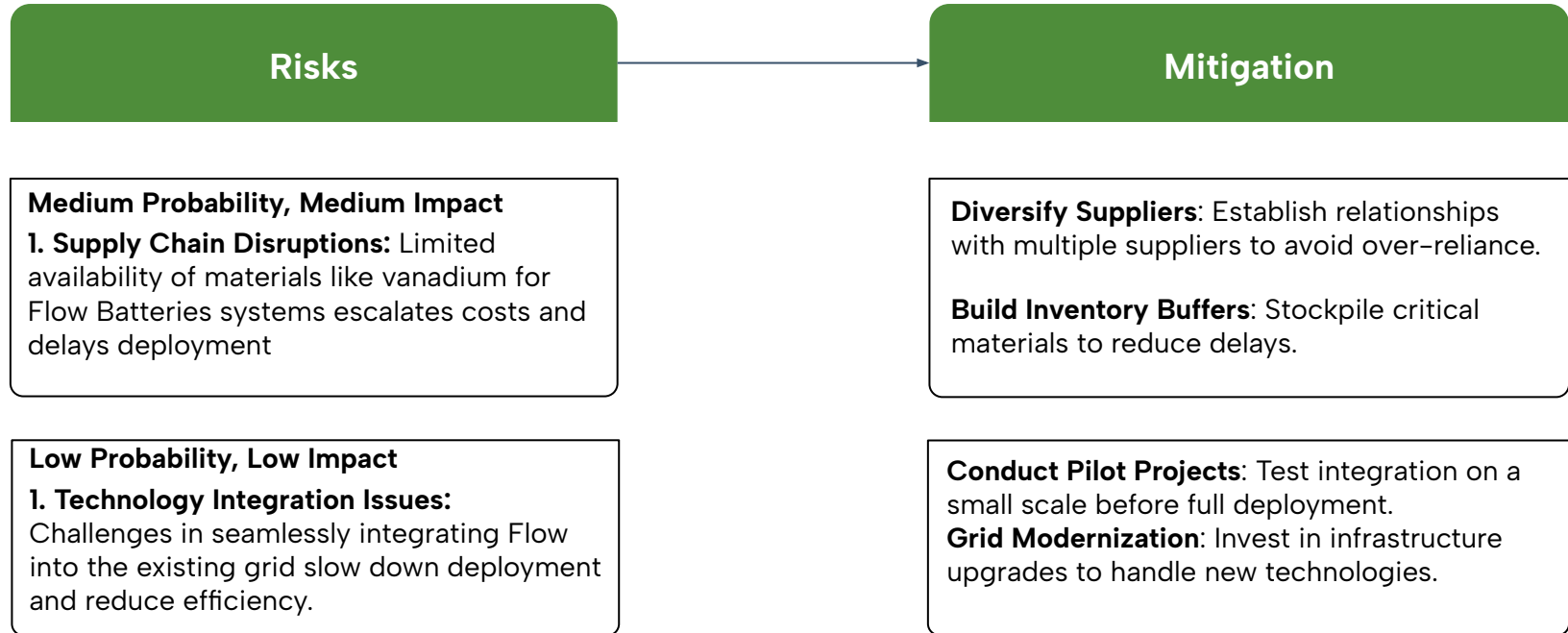
CAPEX- Left over goal - 900 MW

=> Capacity = Power (MW) $\times$ Storage Duration = $900,000 \times 6 \times 125$

Evaluating risks through an analysis of impact and probability criteria, and proposing mitigation strategies



Evaluating risks through an analysis of impact and probability criteria, and proposing mitigation strategies



Phasing out the use of Li-Ion gradually while shifting toward Flow would need transition/adoption curve, accounting for a 30% buffer requirement for its upcoming goals

Enhanced infrastructure for deployment:

- Require larger spaces due to design, including electrolyte storage tanks.

Retraining workforce:

- Require updated training for engineers, technicians, and energy professionals to manage and optimize systems.

Supplier diversification:

- Build manufacturer partnerships specializing in flow battery technology for reliable supply chains.

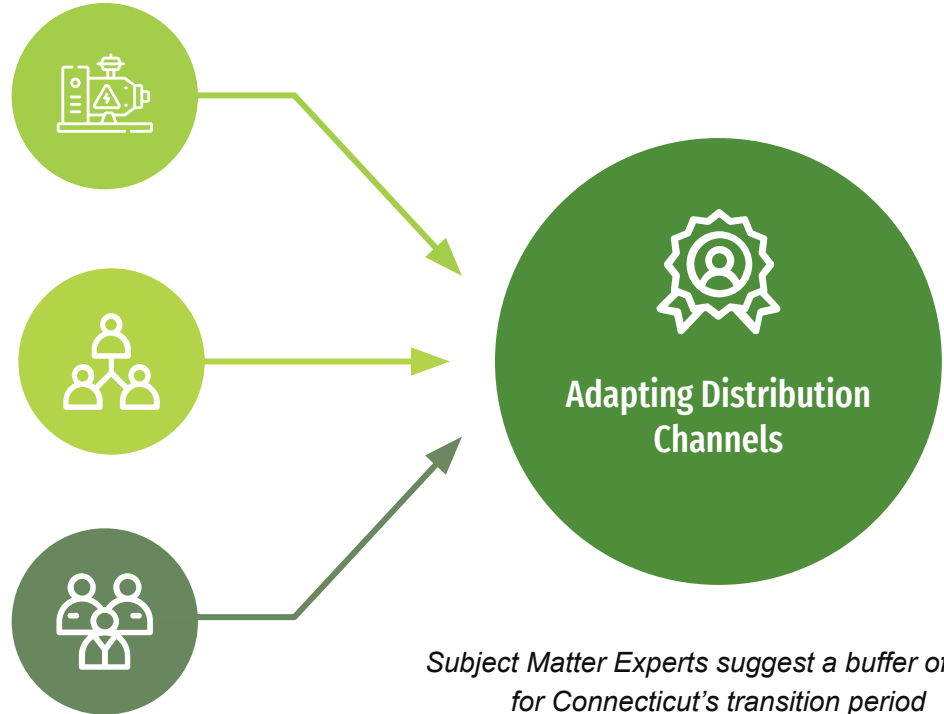


Table 1: Deployment Rates and Milestones

Period	Target MW	Current MW	Required Rate (MW/year)	Adjusted Rate with 20% Delays (MW/year)
2024–2025	150	5.02	72.5	90.6
2026–2027	350	150	100	125
2028–2029	550	350	100	125
2030–2031	700	550	75	93.8

Table 2: Cost Analysis of Lithium-Ion Deployment

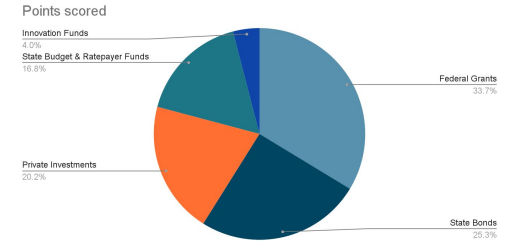
Parameter	Value
Required MW	995
Cost per kWh	\$589
Storage Duration (hours)	4
Total Cost (billion)	\$2.35

Table 3: Comparative State Progress

State	Target MW	Achieved MW	Achievement Rate (%)
Connecticut	1,000	5.02	0.5%
California	1,300	1,533	118%

Table 4: Funding Overview for 700 MW Target

Source	Amount (\$ Billion)
Federal Grants	2.00
State Bonds	1.50
Private Investments	1.20
State Budget & Ratepayer Funds	1.00
Innovation Funds	0.24
Total	5.94



Summary Table for Milestones

Period	Target MW	Required Rate (MW/year)
2024–2025	150	72.5
2026–2027	350	100
2028–2029	550	100
2030–2031	700	75

Our Team's Next Steps Between the Interim and Final Presentation

1. Conduct Cost Analysis: Evaluate total ownership costs (Li-Ion, Flow, Iron-Air) including installation, maintenance, and scalability.
2. Refine Technology Evaluation: Assess performance, scalability, and environmental impact specific to Connecticut
3. Develop Roadmap: Create a timeline for transitioning from Li-Ion to Flow Batteries, aligned with 2027-2030 goals.
4. Present Final Recommendations: Deliver a comprehensive recommendation backed by cost analysis and feasibility studies.

Roadmap: Key Milestones and Timelines for Technology Deployment

Technology	Upfront Cost (\$/kWh)	Lifetime Cost	Lifespan	Technology	Upfront Cost (\$/kWh)	Lifetime Cost	Lifespan
Flow Batteries	\$750 - \$4,000	Lower	25+ years	Flow Batteries	\$750 - \$4,000	Lower	25+ years
Lithium-Ion Batteries	Currently higher, but projected to drop below \$100	Higher due to replacement	3-5 years	Lithium-Ion Batteries	Currently higher, but projected to drop below \$100	Higher due to replacement	3-5 years

Cost-Benefit Analysis: Financial Viability of Selected Technologies

Technology	Initial Cost	Maintenance	Duration	Best Use Case	Financial Viability
Li-Ion	\$300–\$400/kWh	Medium (frequent replacements) <i>2000–3000 charge cycles</i>	Up to 8 hours	Short-term, immediate deployment	Good for short-term goals, but costly long-term
Flow Batteries	\$500–\$700/kWh	Low <i>20,000 – 30,000 charge cycles</i>	10+ hours	Long-duration, grid-scale storage	Cost-effective for long-term needs
Iron–Air Batteries	\$20–\$30/kWh (R&D estimates)	Very low	100+ hours	Ultra-long-duration storage	Promising, but still in development